

Overview of High-Power Wireless Charging Systems and Analysis of Polyphase Wireless Charging System Phase Winding and Resonant Tuning Network Connection Configurations

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Abstract—In spite of the high energy efficiency and environmental benefits of electric vehicles (EVs), adoption rates are increasing at a relatively slow pace, primarily because of EVs' limited range and long charging durations. To reduce EV charging times to be comparable to refueling times of conventional vehicles, extreme fast charging (XFC) systems are required. Such charging systems would reduce drivers' range anxiety and enable long-distance interstate travel with EVs. This substantial target in charging rates with 15–20 min of recharging times requires research and development from grid to batteries with advanced charging systems. Wireless power transfer (WPT) systems for EV charging applications are flexible, convenient, and highly efficient, and they allow automated charging. This article reviews the power electronics and winding and resonant tuning network configurations for polyphase WPT systems for high-power wireless charging applications.

Index Terms—Electric vehicle (EV) charging, extreme fast charging (XFC), wireless power transfer (WPT).

I. INTRODUCTION

REDUCING emissions, increasing collective fuel economy, and decreasing energy consumption across transportation systems are imperative for national security and energy independence [1], [2]. To this end, electrification of passenger vehicles is of extreme importance because of their large share within the national fleet, corresponding to more than 60% of the total transportation sector energy consumption in the United States, electric vehicles (EVs) have great potential to reduce petroleum consumption and greenhouse gas emissions with their inherently high efficiency compared with conventional vehicles [3], [4], [5]. The system-level benefits of the EV ecosystem will grow with increased penetration of renewable energy resources feeding the grid [6], [7]. EVs,

furthermore, significantly support the penetration of smart mobility technologies, most notably connected and automated vehicles [8], [9], [10]. Range anxiety and long charging times are prominent barriers to reducing the further market penetration of electrified mobility solutions. Not having the ability to refuel or recharge quickly is the primary reason for consumers' hesitation to use EVs [11], [12], [13]. High-power charging stations must, therefore, be on par with conventional vehicle refueling practices [14], [15].

Extreme fast charging (XFC) is one of the most important research topics in the field of electromobility, with the potential to significantly reduce charging times [16], [17], [18]. With XFC (i.e., charge rates (C-rate)¹ close to 3C), EV charging times could be reduced to 10 min for a 50% increase in the battery state-of-charge (SOC) [19]. Business trends in the EV market [20], [21], [22] indicate that some original equipment manufacturers are manufacturing EVs with 30–40 kWh battery packs. For these vehicles, a 3C charge rate corresponds to about 100 kW charge power. On the other hand, there is an emerging trend among automotive manufacturers to make EVs with much larger battery capacities and increased range of over 300 miles. For these vehicles, the battery capacity is typically 90–100 kWh, and the 3C charge rates correspond to 270–300 kW charge power.

Additionally, a new trend uses higher voltage battery packs, such as 600–800 V battery packs, to reduce the charge current. Higher battery voltage levels reduce not only the charge current but also the battery current to the traction drive inverters, resulting in higher efficiency in chargers and electric drive systems. The benefits of increased battery voltage in EV systems are outlined in [23] and [24].

With the recent developments in wide bandgap power semiconductor technology, high-frequency and high-power

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¹C-rate is the unit that expresses the charge power/current needed to fully charge or discharge a battery pack under ideal conditions in 1 h. 1C corresponds to the charge/discharge power needed to fully charge or discharge a battery in 1 h. Higher C rates indicate the proportionally reduced time durations. For instance, 2C means 1/2 h needed to fully charge or discharge a battery or 3C means 1/3 h needed to fully charge or discharge a battery.

applications have been able to advance simultaneously [25], [26], enabling higher power wireless power transfer (WPT) systems [27], [28], [29], while 20-kW used to be considered high-power until recently [30]. At high power levels, the required dc cabling and connectors in plug-in EV chargers are too heavy and stiff to physically maneuver and are, therefore, prohibitive for consumer applications [31]. Although integrating liquid cooling for the wire and plug could be considered, this option would add reliability concerns due to possible leaks and additional insulation and periodic maintenance requirements at the charging stations [32]. These liquid-cooled cabling systems, moreover, typically used fan-cooled radiators without refrigeration functionality; therefore, they cannot cool the cable below the ambient temperature, and the cooling performance is limited at elevated ambient temperatures. The other issue with these systems is that wait times may be required between charging sessions. Automated plug systems [33] are a possible solution for high-power charging systems; however, they have not been commercialized because of an excessive number of independently moving joints, actuators, and servomotors that decrease the system's reliability. Overhead charging technologies [34] are used by some manufacturers for buses; however, drawbacks include limited charge power due to the high temperature on contact blades if blade surfaces are not fully aligned (as a result of increased contact resistance) or dust, dirt, moisture, and grease accumulating on the contact surfaces. Conversely, autonomous vehicles cannot reach their full potential if the charging process is not freed from these issues; therefore, the WPT charging approach offers potential advantages to EVs and autonomous vehicles.

There are ongoing research efforts on high-power WPT technologies with some demonstrations and deployments; however, not all these systems exist today or have currently available data. For instance, Utah-based WAVE has a 500 kW WPT project with future demonstration plans [35], while there is limited information about this technology's overall efficiency, topology, complexity, and component count. The Korean Railroad Research Institute developed a 1-MW rated WPT system [36]; however, their technology is more suitable for dynamic wireless charging applications with a very long transmitter (Tx) loop (128 m), which is not practical for light-duty passenger vehicle stationary charging systems. Conductix-Wampfler installed a 60 kW WPT charger for buses in 2003, but limited information is available on their technology, efficiency, coupler dimensions, etc. [37]. IPT Group took over the Primove WPT technology from Bombardier, but there is not extensive technical data available from their new system [38]; furthermore, InductEV (formerly Momentum Dynamics) is working on high-power WPT charging systems based on 75-kW building blocks that can be scaled up for higher power systems with multiple units in parallel based on the limited information available on their website [39]. Recently, Oak Ridge National Laboratory (ORNL) invented [40] a polyphase WPT system [41] that has significant superiorities compared to existing circular, rectangular, square, or double-D type of coupler geometries in terms of surface

TABLE I
SURFACE POWER DENSITY COMPARISONS OF POLYPHASE COILS WITH CONVENTIONAL SYSTEMS

Developer	Power Level [kW]	Coil Dimensions [mm]	Power Density [kW/m ²]
Brusa [51]	3.3	343.90×250	38.39
Bosch [52]	7	260×260	103.55
WiTriCity [53]	11	420×420	62.35
ISE ^I [54]	11	800×800	17
INTIS ^{II} [55]	11	420×300	87
KAIST/WiPowerOne [56]	22	420×560	93.53
HEVO/NYU [57]	25	600×900	46
Toshiba [58]	44 (2×22)	400×300 (ea)	183.33
ETH Zurich [59]	50	410×760	160
Momentum Dynamics ^{III} [60]	65	600×600	180
WAVE IPT [61]	250	1500×1500	111.11
KRRI ^{IV} [62]	250	1400×530	337
ORNL 100 kW polyphase [63]	100	375 (dia)	905.4
ORNL 300 kW polyphase [64]	300	500 (dia)	1530

^I ISE: Institute for Solar Energy Systems

^{II} INTIS: Integrated Infrastructure Solutions

^{III} Now InductEV

^{IV} Korean Railroad Research Institute

TABLE II
WEIGHT AND SPECIFIC POWER COMPARISONS OF POLYPHASE COILS WITH EXISTING SYSTEMS

Developer	Power [kW]	Weight [kg]	Specific Power [kW/kg]
Bosch [52]	7	1.8	0.26
Brusa [65]	22	11.57	1.9
Bombardier [66]	200	340	1.7
WAVE IPT [61]	250	227	0.908
ORNL 100 kW polyphase [63]	100	14	7.14
ORNL 300 kW polyphase [64]	300	30	10

power density (kW/m²) and compactness, and specific power (kW/kg). Some of earlier ORNL publications on polyphase WPT systems include a 50-kW *LCC-LCC* tuned three-phase wireless charging system [42], thermal analysis [43], thermal design, and optimization polyphase WPT systems [44], variable duty cycle control of three-phase voltage source inverters for polyphase WPT systems [45], and how the phase shift duty cycle control can be used for balancing the secondary coil phase currents [46]. Additionally, a shield design for a 50-kW polyphase WPT system is proposed in [47], and interoperability between polyphase and single-phase WPT systems is studied in [48]. Modeling and analysis of polyphase WPT systems are presented in [49], while [50] demonstrates a 100-kW polyphase WPT system experimental test results. The surface power density (kW/m²) comparisons of existing systems and ORNL's polyphase coils are provided in Table I. In Table II, weight and specific power (kW/kg) comparisons are provided.

Challenges to establishing wireless XFC systems include logistics and infrastructure requirements, design and deployment of the grid interface converters, power quality issues, such as grid power factor and total harmonic distortions, availability of grid power and capability to provide

intermittent high-power demand, the potential need for integration with renewable energy or energy storage systems, and distribution voltage level at the point of grid connection. This study focuses on the US Department of Energy's (DOE's) ORNL's experiences and current investigations on high-power wireless fast charging systems, including the power electronics hardware aspects, power converter architectures and their electrical component characteristics, resonant tuning/compensation circuits, electromagnetic couplers, and voltage/current stresses. The rest of this article is organized as follows: Section II-A presents XFC topologies; Section II-B details the series/parallel connected three-phase WPT system configurations and three-phase WPT chargers with star and delta connected phase windings and resonant tuning networks; Section III presents tradeoff analysis and comparisons of these connection configurations, as well as simulation studies and experimental results, and Section IV outlines concluding remarks.

II. EXTREME FAST CHARGER TOPOLOGIES

To satisfy the requirements on XFC systems that provides 50% SOC increase in 10 min, the charger should supply about 300 kW for a vehicle with around 100 kWh of energy storage capacity. Achieving this high-power level for an EV charging system requires collectively designing power electronic converters and electromagnetic systems through consideration of different power architectures.

Parallel connection of converter structures is one of the solutions to reduce the line and converter current to address the drawbacks of conventional charging systems in high-power applications [67], [68], [69]. The current is divided between parallel connected converters that help reduce power losses in the system. Another solution is to use a medium-voltage (MV) level supply where the converters are connected in series or cascaded combinations to reach the MV distribution voltage levels at the front end [70], [71], [72], [73]. Because of the inversely proportional relationship between voltage and current under a constant power level, higher voltage levels result in lower current ratings with lower power losses, which are the dominant loss factor in high-power applications. Development of wide bandgap devices and batteries with higher voltages further accelerates the active development of MV voltage connected charging system topologies for the realization of XFC systems [74], [75], [76].

This section provides an analysis of the state-of-the-art three-phase series/parallel converter structures, which are applicable to XFC systems considering the charge rates of 300 kW for WPT applications. Primary emphases are on power losses; therefore, our discussion focuses on their component level electrical characteristics. In addition, series and *LCC* (series inductor, parallel capacitor, series capacitor) resonant compensation circuits are investigated with three-phase couplers considering star-delta ($Y - \Delta$) connection. Potential future studies considering series and parallel connection of the converters are presented in Section II-C.

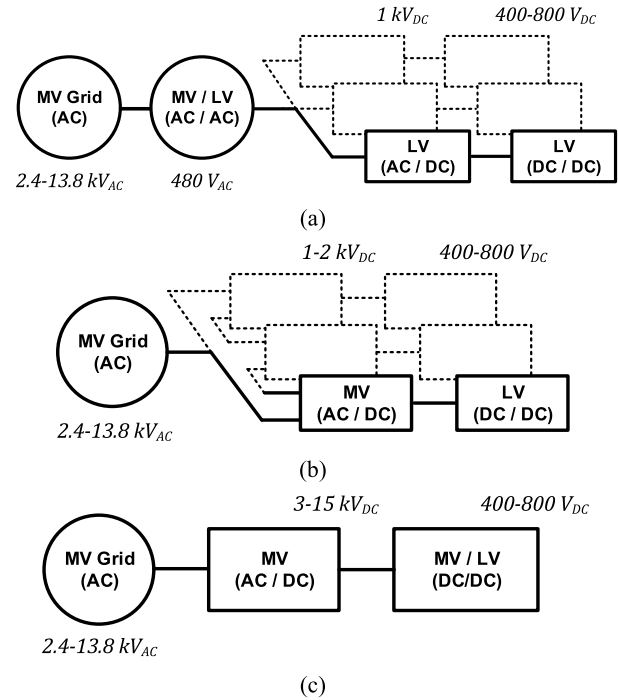


Fig. 1. Overview of MV connected power transfer XFC architectures: (a) parallel connection of ac/dc and dc/dc conversion from the line-frequency transformer, (b) series combination of the ac/dc and dc/dc conversion from MV grid connection, and (c) direct ac/dc and dc/dc conversion using high-voltage rated wide bandgap devices from MV grid connection.

A. High-Power Three-Phase AC/DC PFC Rectifiers and DC/DC WPT Converters

The overview of conventional high-power charger system architectures is illustrated in Fig. 1. Here, MV refers to distribution level voltages, the most common of which are 2.4, 4.2, 6.9, 7.2, 12, and 13.8 kV. The low voltage (LV) is 480 V three-phase at the input of the charging system. The general concept of MV grid voltage to LV conversion by a low-frequency (LF) transformer is shown in Fig. 1(a); here, the parallel connection of ac/dc and dc/dc converters allows access to higher power levels. For instance, there may be three 100 kW units or six 50 kW units in parallel for a 300-kW system. The ac/dc conversion is typically an active rectifier with power factor correction (PFC). The dc/dc stage can be a dual active bridge with a closely coupled isolation transformer or a high-frequency inverter, a resonant stage with loosely coupled WPT coils and their tuning/compensation network, and a high-frequency rectifier. Two other MV to LV methods are given in Fig. 1(b) and (c). In Fig. 1(b), the rectifier is formed by a series or cascaded connected group of rectifiers with each subgroup of rectifiers providing LV (400–800 V) dc to the dc/dc converters. In Fig. 1(c), high-voltage rated wide bandgap devices can be connected directly to the MV line for direct rectification of MV line voltage.

Galvanic isolation enhances the safety of the system and isolates the grid from the load, providing the output is floating with respect to the input; moreover, isolation prevents multiple ground loops and eliminates the risk of current flowing through a person's body to the Earth if there is a fault or if the user

touches a live point in the system on the load side. Galvanic isolation is a safety requirement by the National Electrical Code and the Underwriters Laboratory and is covered in standards such as IEC60950 and UL2202. In MV connected systems, isolation usually is provided at the high-frequency link because high-frequency transformers can be very small and lightweight compared with 60 Hz transformers. In a WPT system, coupling coils with large magnetic airgap separation provide galvanic isolation while also providing the coupling between the ground and vehicle side assemblies. In a direct MV connected system with very high-voltage rated devices, voltage reduction may be achieved by adjusting the number of turns on the primary and secondary sides of the coupling coils and by designing a small resonant voltage gain (<1) when selecting the resonant tuning network parameters. On the other hand, based on IEEE C57.12.00-2010 [77], minimum clearance is 6.5 in between live parts of different phases for a 15 kV power transformer; thus, series combinations of dc/dc converter stages might be required after the ac/dc stage, or MV to LV conversion might be needed before the dc/dc stage. As a result, series and parallel combinations of power converters are crucial for high-power WPT charging systems.

Most of the widely used XFC architectures use an LF transformer to convert MV to LV at the front-end stage; this is followed by an LV (480 V three-phase) connected active rectifier or a diode bridge rectifier, which is then followed by a dc/dc converter to accomplish PFC and primary side dc bus voltage regulation [78], [79], [80]. The two-level rectifiers have the advantage of design and control simplicity, high robustness, and wide compatibility with existing charging systems. For high-power applications, the two-level converter topologies have a definite limitation of maximum power. One way to increase the power ratings of the topologies while maintaining positive properties is to parallel a number of devices to establish phase-shifted multiinterleaved converters. This expansion provides many advantages, such as reducing the size of the electromagnetic interference filter and the output filter of the system.

Multilevel converter topologies have been investigated for high-input voltage and high-power applications [81], [82], [83]. The multilevel converter topologies generate multiple output voltage levels (three or more levels) centered at the neutral node voltage. Advantages of the multilevel converter include reduced blocking voltage stress for the power devices, resulting in lower switching and conduction losses and reduced volume of passive components. In addition, multilevel converters improve the power quality on the grid side with reduced total harmonic distortions. On the other hand, the converter type has high complexity in control and hardware design.

The concepts of two- and three-level central dc/dc converters, including conventional three-phase, neutral point-clamped (NPC), and flying-capacitor (FC) converters, have been proposed as promising solutions for high-power systems [84], [85], [86]. The three-level NPC converter is widely used for various high-power applications with relatively high input voltage. One potential issue that must be considered in an NPC converter is that there may be a voltage imbalance problem on the dc-bus with nonaccurate midpoint voltage regulation.

Although FC converter topology can achieve better voltage balancing, specific control or circuitry is required to pre-charge capacitors, which results in high control complexity; moreover, the FC converter requires more dc bus capacitance compared with the NPC converter, and thus, usually has a larger physical size and weight than the NPC converter. The cascaded three-phase converter requires minimum effort, but this type of converter requires isolation in each phase that prevents connecting the three-phase WPT coupling coils in star and delta configurations.

With single-phase WPT coupling coils including circular and double-D type couplers, a single-phase high-frequency inverter on the primary side and a single-phase rectifier on the secondary side are commonly used. Recently, ORNL outlined the limitations of single-phase WPT systems and the benefits of polyphase WPT systems using rotating fields [40], [41]. For higher power applications, polyphase WPT systems offer significantly higher surface power densities without needing multiple couplers on ground and vehicle assemblies. High-power three-phase WPT systems with dc input and dc output are shown in Fig. 2. The dc/dc converters shown in Fig. 1 can use one of these systems shown in Fig. 2 for a wireless charging system while conventional dual active bridge systems could be used for plug-in charging applications.

B. Series/Parallel Three-Phase WPT System

In this section, three-phase WPT system with “input series/output parallel” and “input parallel/output parallel” connections are discussed, with consideration given to series and LCC resonant tuning circuits. Star or wye (Y) and delta (Δ) connection types for the three-phase couplers are, furthermore, analyzed and compared for voltage and current amplitudes in Section II-C.

C. Series and Parallel Three-Phase WPT System

“Input series/output parallel” and “input parallel/output parallel” three-phase connection configurations are shown in Fig. 3. As seen from Fig. 3(a), based on the output of PFC voltage amplitude, dc bus voltage is divided into the number of series connected converters. With this connection scheme, the input dc voltage amplitude can be high and proportional to each converter’s input dc voltage amplitude. For the input parallel connection, the input current can be shared through the parallel connection of three-phase converters, as shown in Fig. 3(b). Whereas the architecture illustrated in Fig. 3(a) is suitable for systems with MV (several kilovolts) connection, the architecture shown in Fig. 3(b) is suitable for conventional 480 V three-phase connected systems with high throughput power.

The series system total input voltage $V_{dc,s}$, current $i_{dc,s}$, and power $P_{dc,s}$ equations can be written as

$$V_{dc,s} = \sum_{u=1}^n V_{Ci,1,\dots,u} \quad (1)$$

$$i_{dc,s} = i_{Ci,1,\dots,n} \quad (2)$$

$$P_{dc,s} = V_{dc,s} i_{dc,s} \quad (3)$$

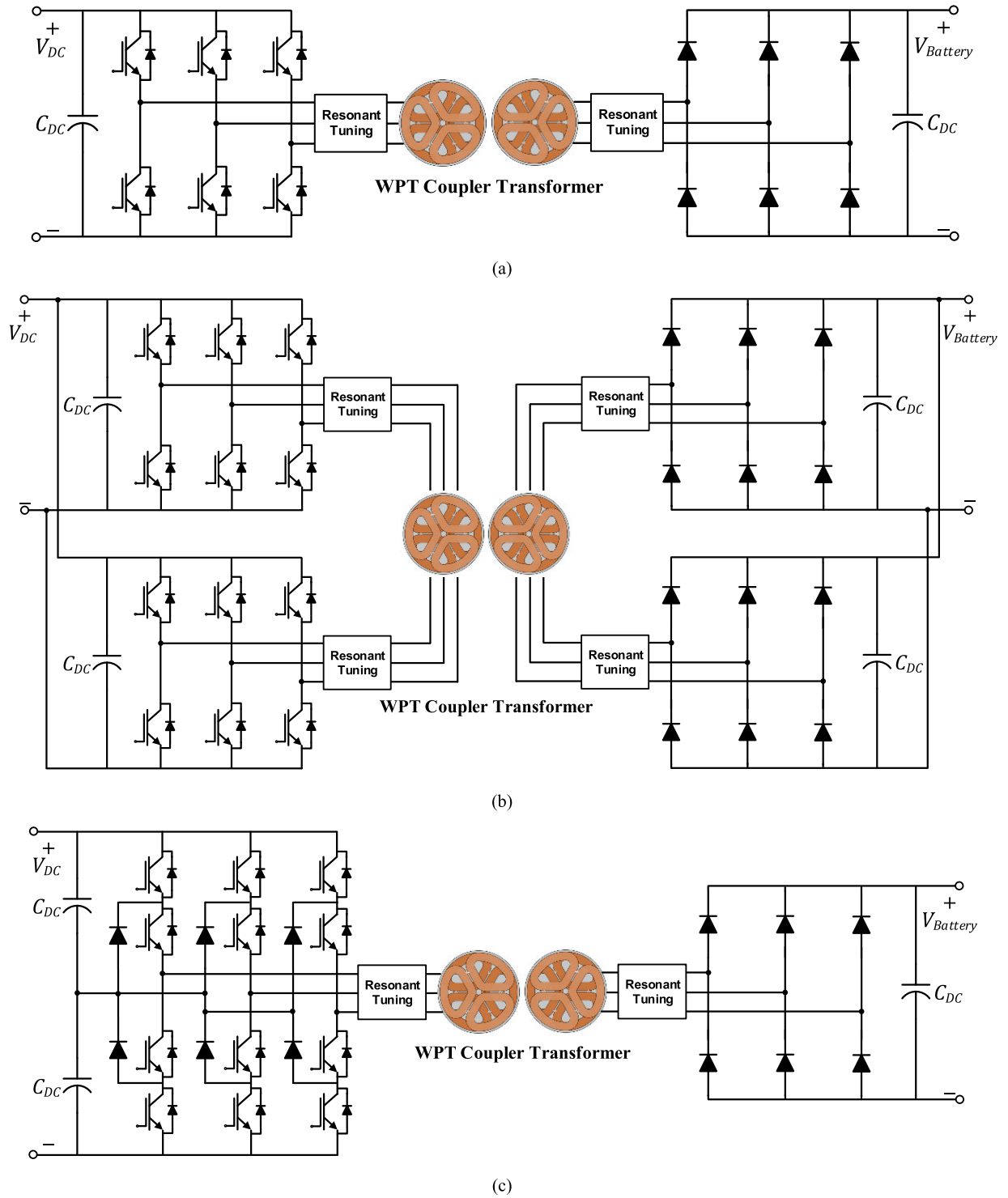


Fig. 2. Typical two- and multilevel dc/dc WPT EV charger topologies: (a) three-phase dc/dc WPT converter, (b) open-ended winding three-phase dc/dc WPT converter, and (c) three-level NPC type dc/dc converter with high-frequency isolation stages.

The parallel system input voltage $V_{dc,p}$, total current $i_{dc,p}$, and power $P_{dc,p}$ calculations can be described as

$$V_{dc,p} = v_{Ci,1,\dots,n} \quad (4)$$

$$i_{dc,p} = \sum_{u=1}^n i_{Ci,u} \quad (5)$$

$$P_{dc,p} = V_{dc,p} i_{dc,p}. \quad (6)$$

The parallel system output battery voltage $V_{battery}$, total current $i_{battery}$, and power $P_{battery}$ equations can be given as

$$V_{battery} = v_{Co,1,\dots,n} \quad (7)$$

$$i_{battery} = \sum_{u=1}^n i_{Co,u} \quad (8)$$

$$P_{battery} = V_{battery} i_{battery}. \quad (9)$$

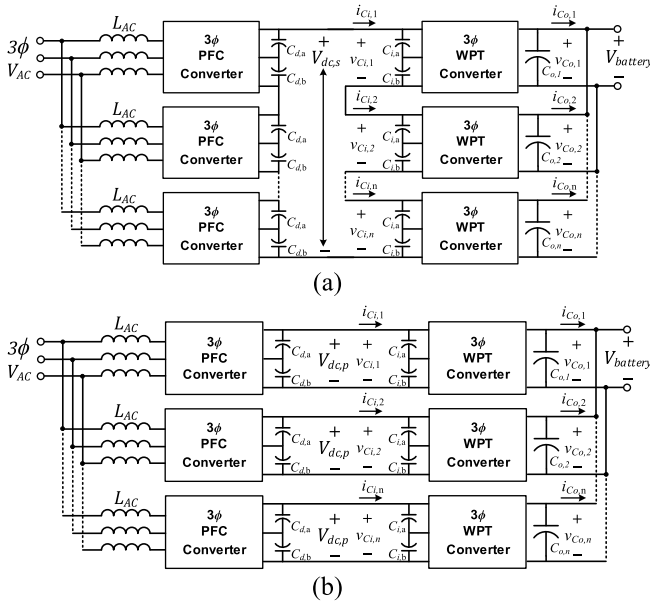


Fig. 3. Three-phase ac/dc and dc/dc converter connection configurations: (a) input series (cascaded) and output parallel and (b) input parallel and output parallel.

III. WPT THREE-PHASE COUPLER STAR (Y) AND DELTA (Δ) CONNECTIONS

Recently, the authors developed polyphase WPT systems using bipolar windings and rotating magnetic fields to significantly improve the power density and specific power of WPT couplers [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [87]. While the single-phase systems, including circular and DD couplers, pulse the power across the airgap between zero and peak, which results in low space-time average use, polyphase systems can be designed to use rotating fields for constant power transfer. In polyphase systems, while the excitation is phase shifted, the phase windings are also spatially shifted, resulting in much higher power densities due to improved space-time field use. With this feature, these couplers can be designed to use the field-shaping nature of the bipolar phase windings to meet field emission limits imposed by safety guidelines while improving power transfer capability. In these systems, the input and output current ripples are significantly lower; therefore, smaller dc bus capacitors are sufficient compared with the large dc bus capacitor requirements of single-phase systems. On the other hand, resonant networks must be tuned properly considering interphase mutual inductances to fully exploit the benefits of polyphase systems with nonzero mutual coupling between phase windings.

Mutually coupled three-phase coupler circuit diagrams are presented with inductance matrix for wye (star) and delta connections in Fig. 4.

The equivalent inductances L_P and L_S can be characterized with inductance matrices as shown by (10)–(12), where L_P denotes the primary-side inductance matrix with self-inductances of the phase windings, and the mutual inductances between the primary-side phase windings and L_S show the secondary-side inductance matrix with self-inductances of

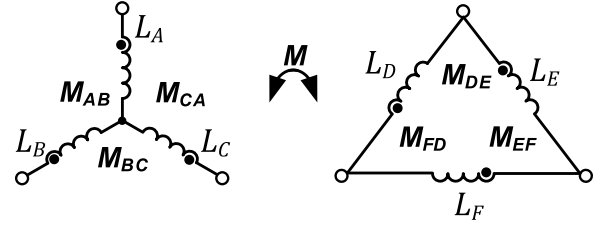


Fig. 4. Mutually coupled three-phase coupler with Y/ Δ connection configuration.

the phase windings and the mutual inductances between the secondary-side phase windings

$$L_P = \begin{bmatrix} L_A & M_{AB} & M_{CA} \\ M_{AB} & L_B & M_{BC} \\ M_{CA} & M_{BC} & L_C \end{bmatrix} \quad (10)$$

$$L_S = \begin{bmatrix} L_D & M_{DE} & M_{FD} \\ M_{DE} & L_E & M_{EF} \\ M_{FD} & M_{EF} & L_F \end{bmatrix}. \quad (11)$$

Because of the cross-couplings between primary and secondary phase windings, the overall inductance matrix of the whole system can be written in a six-by-six matrix form as follows:

$$L = \begin{bmatrix} L_A & M_{AB} & M_{AC} & M_{AD} & M_{AE} & M_{AF} \\ M_{BA} & L_B & M_{BC} & M_{BD} & M_{BE} & M_{BF} \\ M_{CA} & M_{CB} & L_C & M_{CD} & M_{CE} & M_{CF} \\ M_{DA} & M_{DB} & M_{DC} & L_D & M_{DE} & M_{DF} \\ M_{EA} & M_{EB} & M_{EC} & M_{ED} & L_E & M_{EF} \\ M_{FA} & M_{FB} & M_{FC} & M_{FD} & M_{FE} & L_F \end{bmatrix}. \quad (12)$$

In this inductance matrix, the mutual couplings between the same phase windings are equal to each other such that $M_{AB} = M_{BA}$, $M_{AC} = M_{CA}$, $M_{AD} = M_{DA}$, $M_{AE} = M_{EA}$, $M_{AF} = M_{FA}$, $M_{BC} = M_{CB}$, $M_{BD} = M_{DB}$, $M_{BE} = M_{EB}$, $M_{BF} = M_{FB}$, $M_{CD} = M_{DC}$, $M_{CE} = M_{EC}$, $M_{CF} = M_{FC}$, $M_{DE} = M_{ED}$, $M_{DF} = M_{FD}$, and $M_{EF} = M_{FE}$. While the representation in Fig. 4 is for a Y– Δ connection configuration, the three-phase couplers of primary and secondary windings can also be connected in Y–Y (star-star), Y– Δ (star-delta), Δ –Y (delta-star), or Δ – Δ (delta-delta) connection configurations as shown in Fig. 5.

The coupler primary/secondary phase voltages $V_{\text{phase}_p}/V_{\text{phase}_s}$ and currents $i_{\text{phase}_p}/i_{\text{phase}_s}$ can be expressed considering input and output phase voltages and currents. For the Y–Y connected coupler transformer, primary and secondary side voltage and current equations can be shown as

$$V_{\text{phase}_p} = \sqrt{3}V_{AN} \quad (13)$$

$$i_{\text{phase}_p} = i_A \quad (14)$$

$$V_{\text{phase}_s} = \sqrt{3}V_{DN'} \quad (15)$$

$$i_{\text{phase}_s} = i_D \quad (16)$$

$$V_{\text{phase}_p} = V_{\text{phase}_s} \sqrt{L_S}/L_P = \sqrt{3}V_{DN'} \sqrt{L_S}/L_P \quad (17)$$

$$i_{\text{phase}_p} = i_{\text{phase}_s} \sqrt{L_P}/L_S = i_D \sqrt{L_P}/L_S. \quad (18)$$

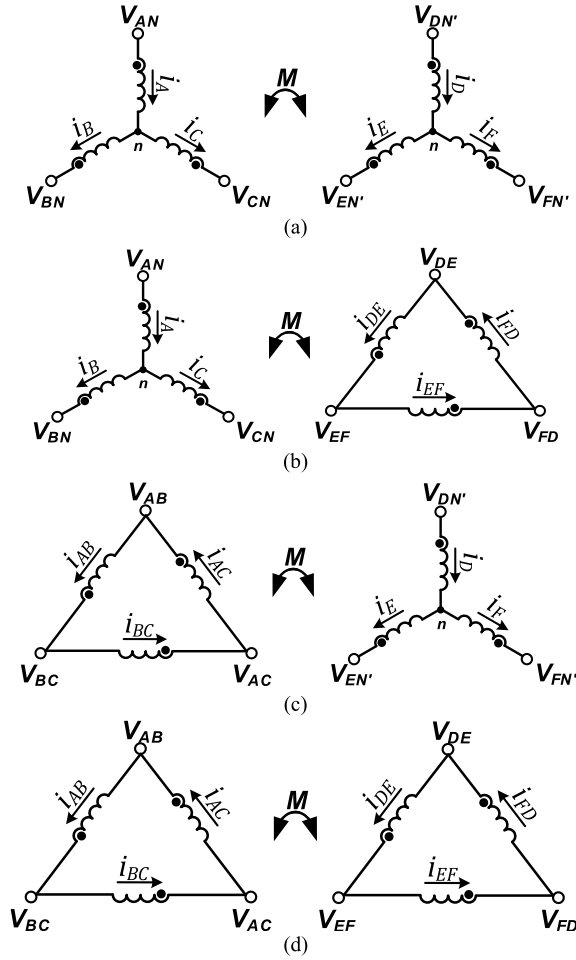


Fig. 5. Three-phase Tx and Rx coupler connection configurations; (a) Y/Y , (b) Y/Δ , (c) Δ/Y , and (d) Δ/Δ .

The primary and secondary side voltage and current equations for the $Y - \Delta$ coupler can be written as

$$V_{\text{phase}_p} = \sqrt{3} V_{AN} \quad (19)$$

$$i_{\text{phase}_p} = i_A \quad (20)$$

$$V_{\text{phase}_s} = \sqrt{3} V_{DE} \quad (21)$$

$$i_{\text{phase}_s} = i_{DE} \quad (22)$$

$$V_{\text{phase}_p} = \frac{V_{\text{phase}_s}}{\sqrt{3}} \sqrt{L_S/L_P} = \frac{V_{DE}}{\sqrt{3}} \sqrt{L_S/L_P} \quad (23)$$

$$i_{\text{phase}_p} = \sqrt{3} i_{\text{phase}_s} \sqrt{L_P/L_S} = 3 i_{DE} \sqrt{L_P/L_S}. \quad (24)$$

The primary and secondary side voltage and current equations for the $\Delta - Y$ coupler design can be written as

$$V_{\text{phase}_p} = V_{AB} \quad (25)$$

$$i_{\text{phase}_p} = \sqrt{3} i_{AB} \quad (26)$$

$$V_{\text{phase}_s} = \sqrt{3} V_{DN'} \quad (27)$$

$$i_{\text{phase}_s} = i_D \quad (28)$$

$$V_{\text{phase}_p} = \sqrt{3} V_{\text{phase}_s} \sqrt{L_S/L_P} = \sqrt{3} V_{DN'} \sqrt{L_S/L_P} \quad (29)$$

$$i_{\text{phase}_p} = \frac{i_{\text{phase}_s}}{\sqrt{3}} \sqrt{L_P/L_S} = \frac{i_D}{\sqrt{3}} \sqrt{L_P/L_S}. \quad (30)$$

For the delta/delta coupler design, the primary and secondary side voltage and current equations can be written as

$$V_{\text{phase}_p} = V_{AB} \quad (31)$$

$$i_{\text{phase}_p} = \sqrt{3} i_{AB} \quad (32)$$

$$V_{\text{phase}_s} = V_{DE} \quad (33)$$

$$i_{\text{phase}_s} = \sqrt{3} i_{DE} \quad (34)$$

$$V_{\text{phase}_p} = V_{\text{phase}_s} \sqrt{L_S/L_P} = V_{DE} \sqrt{L_S/L_P} \quad (35)$$

$$i_{\text{phase}_p} = i_{\text{phase}_s} \sqrt{L_P/L_S} = \sqrt{3} i_{DE} \sqrt{L_P/L_S}. \quad (36)$$

As seen from the equations, a delta-delta ($\Delta - \Delta$) connection provides less current stress in the coupler primary and secondary windings.

IV. SERIES AND LCC RESONANT COMPENSATION CIRCUITS

The most commonly used resonant compensation circuits are the series and LCC resonant tuning networks, whereas series tuning offers simplicity and cost effectiveness, LCC provides advanced functionalities such as load and coupling factor independent constant current source behavior on the primary coupler. With that feature, a cycle-by-cycle primary-side control requirement is eliminated, and the communication system between primary and secondary sides can be simplified; i.e., the need for a very fast control system is eliminated because the primary side operates with constant coil current regardless of the load and coupling conditions. In addition, LCC tuning can reduce inverter and rectifier component current stresses for improved performance.

Three-phase series and LCC compensation circuit schematics are given for the three-phase WPT coupler star and delta connections in Figs. 6 and 7, respectively. The circuit resonant compensation calculations are also presented for each of the compensation configurations. As seen from the expressions, each system's resonant compensation depends on the coupler and resonant tuning network connection types, which also determines the system voltage/current gain margins.

For series resonant compensation circuit design, series resonant capacitors C_{SS} , C_{DS} , C_{SD} , and C_{DD} can be calculated for star/star (Y/Y), star/delta (Y/Δ), delta/star (Δ/Y), and delta/delta (Δ/Δ) as

$$C_{SSa} = C_{SSb} = C_{SSc} = \frac{1}{4\pi^2 f_{sw}^2 L_P}, \quad \text{for } Y - Y \quad (37)$$

$$C_{DSa} = C_{DSb} = C_{DSc} = \frac{1}{12\pi^2 f_{sw}^2 L_P}, \quad \text{for } Y - \Delta \quad (38)$$

$$C_{SDa} = C_{SDb} = C_{SDc} = \frac{3}{4\pi^2 f_{sw}^2 L_P}, \quad \text{for } \Delta - Y \quad (39)$$

$$C_{DDa} = C_{DDb} = C_{DDc} = \frac{1}{4\pi^2 f_{sw}^2 L_P}, \quad \text{for } \Delta - \Delta. \quad (40)$$

It should be noted that whereas the primary side can have one of the connection configurations shown in Fig. 6(a)–(d), the secondary side may have the same or a different connection configuration. This results in 16 different coupler and tuning capacitor connection configurations (combinations of $Y - Y$, $Y - \Delta$, $\Delta - Y$, and $\Delta - \Delta$) for the series-series tuned three-phase WPT systems.

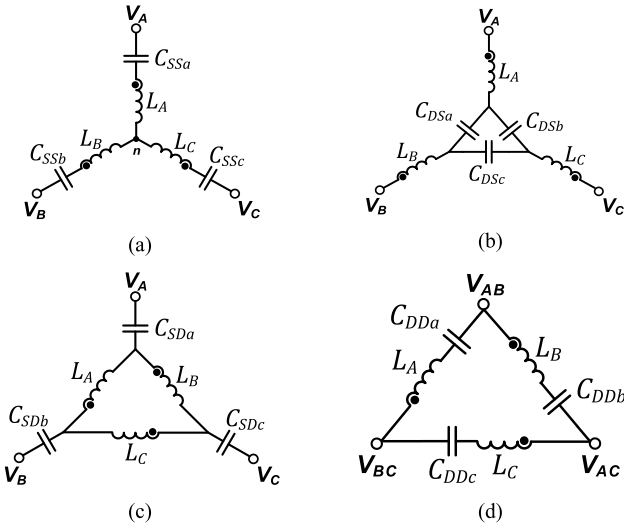


Fig. 6. Three-phase resonant compensation circuits series-series tuning for (a) coupling inductor Y and series capacitor Y ($Y-Y$), (b) coupling inductor Y and series capacitor Δ ($Y-\Delta$), (c) coupling inductor Δ and series capacitor Y ($\Delta-Y$), and (d) coupling inductor Δ and series capacitor Δ ($\Delta-\Delta$).

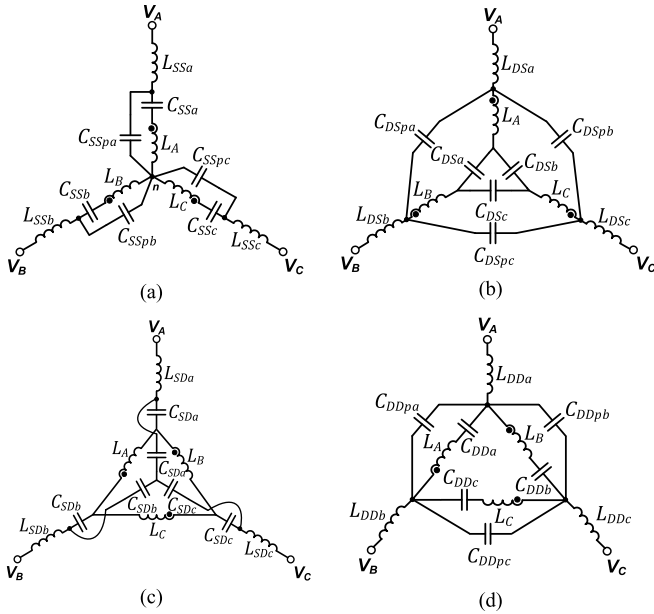


Fig. 7. Three-phase resonant compensation $LCC-LCC$ tuning for (a) coupling inductor star and series and parallel capacitors star ($Y-Y$), (b) coupling inductor star and series and parallel capacitors delta ($Y-\Delta$), (c) coupling inductor delta and series and parallel capacitors star ($\Delta-Y$), and (d) coupling inductor delta and series and parallel capacitors delta ($\Delta-\Delta$).

For the LCC resonant tuning network, similar to the series-series tuned system, the series and parallel capacitors can be connected in delta and star configurations, whereas the coupler windings can also be connected in star or delta configurations, as shown in Fig. 7. In all of these connection configurations, the tuning inductor is in series with the rest of the components, similar to the single-phase LCC tuned systems. Similar to the series-series tuned three-phase systems, primary and secondary sides can have a combination of star and delta-type connection configurations like those described above (combination of Y/Y , Y/Δ , Δ/Y , Δ/Δ).

For LCC resonant compensation circuit design, series resonant capacitors C_{SS} , C_{DS} , C_{SD} , C_{DD} ; parallel resonant capacitors C_{SSp} , C_{DSP} , C_{SDp} , C_{DDp} , and series inductors L_{SS} , L_{DS} , L_{SD} , L_{DD} can be calculated for star-star ($Y-Y$), star-delta ($Y-\Delta$), delta-star ($\Delta-Y$), and delta-delta ($\Delta-\Delta$) as

$$C_{SSa} = C_{SSb} = C_{SSc} = \frac{1}{4\pi^2 f_{sw}^2 (L_P - L_{SSa})}$$

$$C_{SSpa} = C_{SSpb} = C_{SSpc} = \frac{1}{4\pi^2 f_{sw}^2 L_{SSa}}$$

$$L_{SSa} = L_{SSb} = L_{SSc} = \frac{2V_{in}}{2\sqrt{2}\pi^2 f_{sw} i_{AR}}, \quad \text{for } Y-Y \quad (41)$$

$$C_{DSa} = C_{DSb} = C_{DSc} = \frac{1}{4\pi^2 f_{sw}^2 (3L_P - 3L_{DSa})}$$

$$C_{DSPa} = C_{DSPb} = C_{DSPc} = \frac{1}{12\pi^2 f_{sw}^2 L_{DSa}}$$

$$L_{DSa} = L_{DSb} = L_{DSc} = \frac{2V_{in}}{2\sqrt{2}\pi^2 f_{sw} i_{AR}}, \quad \text{for } Y-\Delta \quad (42)$$

$$C_{SDa} = C_{SDb} = C_{SDc} = \frac{1}{4\pi^2 f_{sw}^2 (\frac{L_P}{3} - L_{SDa})}$$

$$C_{SDpa} = C_{SDpb} = C_{SDpc} = \frac{1}{4\pi^2 f_{sw}^2 L_{SDa}}$$

$$L_{SDa} = L_{SDb} = L_{SDc} = \frac{2V_{in}}{2\sqrt{2}\pi^2 f_{sw} i_{AR}}, \quad \text{for } \Delta-Y \quad (43)$$

$$C_{DDa} = C_{DDb} = C_{DDc} = \frac{1}{4\pi^2 f_{sw}^2 (L_P - 3L_{DDa})}$$

$$C_{DDpa} = C_{DDpb} = C_{DDpc} = \frac{1}{12\pi^2 f_{sw}^2 L_{DDa}}$$

$$L_{DDa} = L_{DDb} = L_{DDc} = \frac{2V_{in}}{2\sqrt{2}\pi^2 f_{sw} i_{AR}}, \quad \text{for } \Delta-\Delta \quad (44)$$

where i_{ar} is the current through the inductors.

V. TRADEOFF ANALYSIS AND DISCUSSIONS

According to (37)–(40), for the series-series tuned three-phase WPT system, the capacitor value is the lowest for the $Y-\Delta$ connected system. Capacitor value alone, however, does not mean much because the capacitor size and cost depend on the voltage and current rating (component stress) of the capacitor. Similarly, for the $LCC-LCC$ tuned system, the voltage and current stress on the series and parallel tuning capacitors are more important than the values of the capacitors. For the tuning inductor, the inductance value remains the same for all connection configurations, as seen in (41)–(44).

As outlined in Table III, with polyphase wireless charging coils, 16 different connection configuration options are available for the system. As an example, $Y-\Delta/\Delta-Y$ indicates that primary coils are connected in Y , the primary resonant tuning network is connected in Δ , secondary coils are connected in Δ , and the secondary resonant network is connected in Y , respectively. As a result, a series-series tuned

TABLE III
COIL AND RESONANT TUNING NETWORK CONNECTION
CONFIGURATION OPTIONS

Connection configuration option #	Primary-side coil and tuning connection configuration	Secondary-side coil and tuning connection configuration
1	Y-Y	Y-Y
2	Y-Y	Y-Δ
3	Y-Y	Δ-Y
4	Y-Y	Δ-Δ
5	Y-Δ	Y-Y
6	Y-Δ	Y-Δ
7	Y-Δ	Δ-Y
8	Y-Δ	Δ-Δ
9	Δ-Y	Y-Y
10	Δ-Y	Y-Δ
11	Δ-Y	Δ-Y
12	Δ-Y	Δ-Δ
13	Δ-Δ	Y-Y
14	Δ-Δ	Y-Δ
15	Δ-Δ	Δ-Y
16	Δ-Δ	Δ-Δ

TABLE IV
SELF AND MUTUAL INDUCTANCES OF PRIMARY
AND SECONDARY PHASE WINDINGS

	Self and mutual inductances																	
Primary-side self and mutual inductances [μH]	L_a		L_b		L_c		M_{ab}		M_{bc}		M_{ca}							
	31.7		31.7		31.7		-10.2		-10.2		-10.2							
Secondary-side self and mutual inductances [μH]	L_x		L_y		L_z		M_{xy}		M_{yz}		M_{zx}							
	31.7		31.7		31.7		-10.2		-10.2		-10.2							
Primary and secondary mutual inductances [μH]	M_{ax}		M_{ay}		M_{az}		M_{bx}		M_{by}		M_{bz}		M_{cx}		M_{cy}		M_{cz}	
	4.9		1.6		3.5		3.5		4.9		1.6		1.6		3.5		4.9	

system would have 16 different coil connections and resonant tuning connection configurations. Similarly, *LCC-LCC* and *LCC-series* systems each would also have 16 different connection configuration combinations possible. This results in 48 different architectures in total. In this section, simulation studies are presented to compare these 16 connection configuration combinations for series-series, *LCC-LCC*, and *LCC-series* connection configurations.

A. Simulation Studies

The three-phase WPT EV charger system is simulated with coupler star and delta combinations for series-series, *LCC-LCC*, and *LCC-series* tuning, as summarized in Figs. 8–10, respectively. The system tradeoff analyses are carried out considering the voltage and current stresses in the passive components. The system input dc bus voltage is 1600 V for conventional multilevel WPT converters, as described in Fig. 2. This dc bus voltage level can be

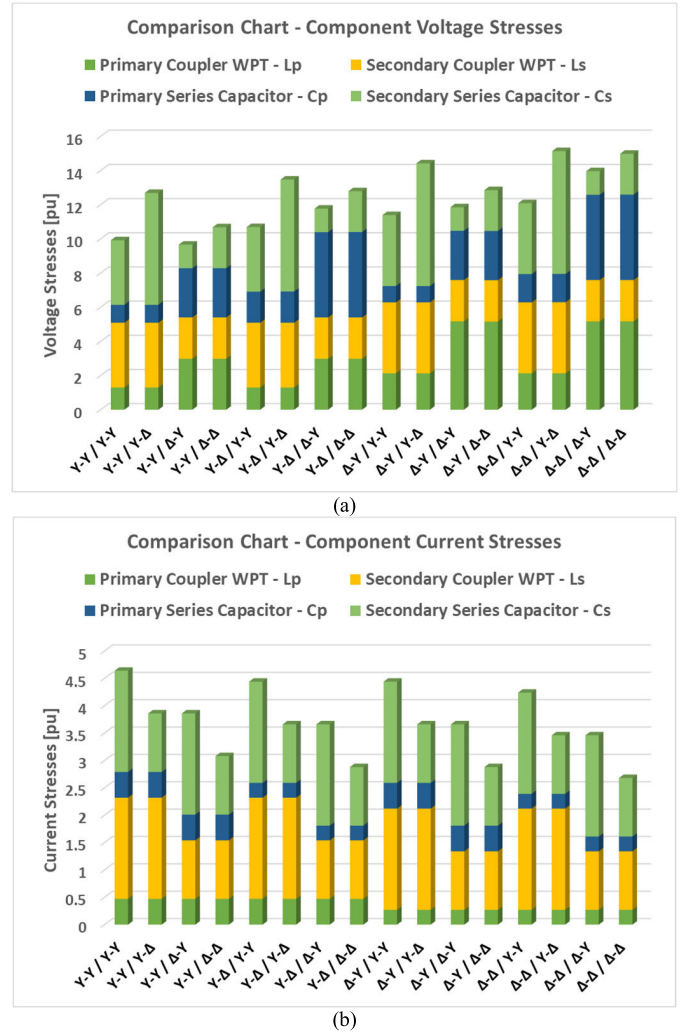


Fig. 8. Three-phase converter with series-series tuning for three-phase coupler star and delta combinations: (a) voltage stresses and (b) current stresses.

TABLE V
RESONANT TUNING COMPONENT VALUES FOR SERIES-SERIES
TUNING CONFIGURATION

Compensation topology	Series-Series			
Connection configuration	Y-Y	Y-Δ	Δ-Y	Δ-Δ
Tuning capacitor	$C_{SSa} = C_{SSb} = C_{SSc}$	$C_{DSa} = C_{DSb} = C_{DSc}$	$C_{SDa} = C_{SDb} = C_{SDc}$	$C_{DDa} = C_{DDb} = C_{DDc}$
Value [nF]	83.65	27.88	250.9	83.65

obtained by the active boost rectification of a 600 or 1000 V secondary distribution system voltage. Considering a typical vehicle battery voltage, the system output voltage is selected as 400 V at 100 kW load; furthermore, figures for voltage and current stresses are shown per unit for 1000 V and 100 A bases, respectively. In Table IV, the self and mutual inductance parameters are provided for the polyphase coupling coils where a , b , and c denote the primary-side phase windings and x , y , and z denote the secondary-side windings. Table V shows the resonant tuning capacitor values for the series-series tuning network for all four connection configurations, whereas

TABLE VI
RESONANT TUNING COMPONENT VALUES FOR *LCC-LCC*
TUNING CONFIGURATION

Compensation topology	<i>LCC-LCC</i>			
Connection configuration	Y-Y	Y-Δ	Δ-Y	Δ-Δ
Tuning capacitor	$C_{SSa} = C_{SSb} = C_{SSc}$	$C_{DSa} = C_{DSb} = C_{DSc}$	$C_{SDa} = C_{SDb} = C_{SDc}$	$C_{DDa} = C_{DDb} = C_{DDc}$
Value [nF]	91.476	30.492	337.62	112.54
Parallel tuning capacitor	$C_{SSpa} = C_{SSpb} = C_{SSpc}$	$C_{DSpa} = C_{DSpb} = C_{DSpc}$	$C_{SDpa} = C_{SDpb} = C_{SDpc}$	$C_{DDpa} = C_{DDpb} = C_{DDpc}$
Value [nF]	977.47	325.82	977.47	325.82
Series tuning inductor	$L_{SSa} = L_{SSb} = L_{SSc}$	$L_{DSa} = L_{DSb} = L_{DSc}$	$L_{SDa} = L_{SDb} = L_{SDc}$	$L_{DDa} = L_{DDb} = L_{DDc}$
Value [μH]	3.587	3.587	3.587	3.587

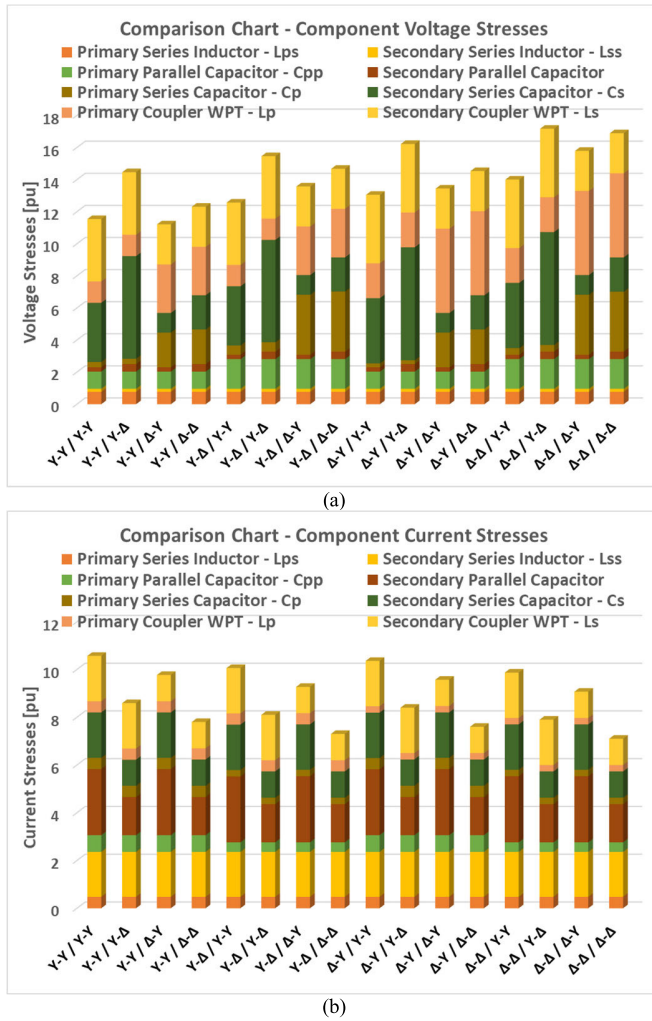


Fig. 9. Three-phase converter with *LCC-LCC* tuning for three-phase coupler star and delta combinations: (a) voltage stresses and (b) current stresses.

Table VI shows the tuning component values for the *LCC-LCC* resonant tuning network. For the *LCC-series* type of tuning network, primary-side resonant tuning components use the

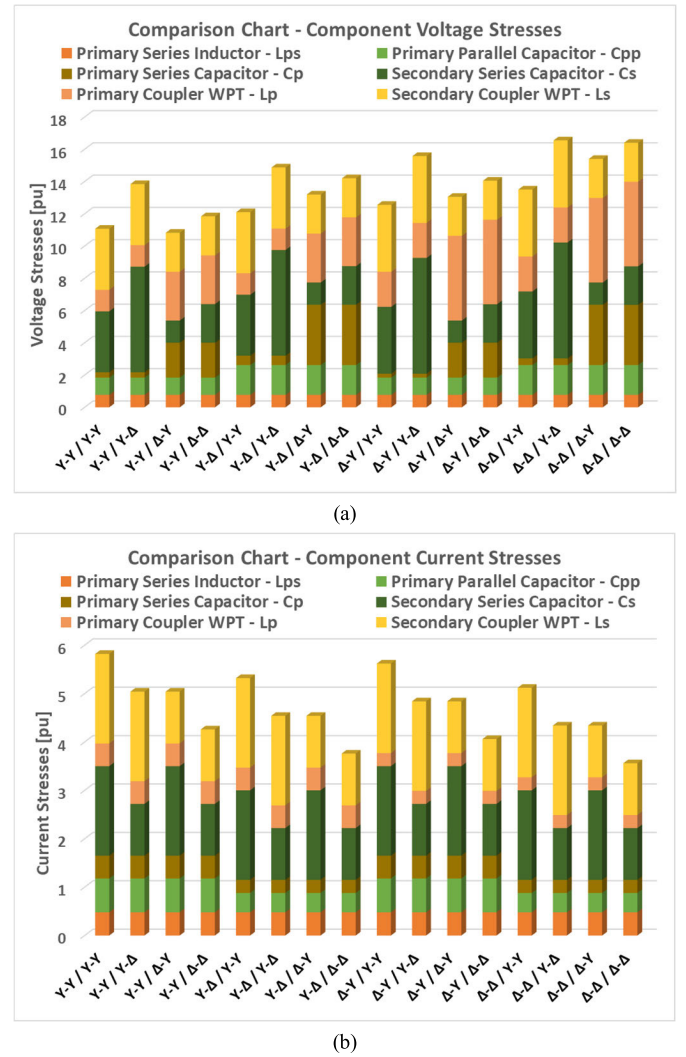


Fig. 10. Three-phase converter with *LCC-series* tuning for three-phase coupler star and delta combinations: (a) voltage stresses and (b) current stresses.

values shown in Table VI, whereas the secondary-side tuning capacitor value is shown in Table V.

In these figures, C_P , C_S , L_P , and L_S refer to the primary-side series tuning capacitor, secondary-side resonant series tuning capacitor, primary-side coupler inductance, and secondary-side coupler inductance, respectively. Additionally, L_{PS} , L_{SS} , C_{PP} , and C_{PS} indicate the primary side series inductor, secondary side series inductor, primary side parallel tuning capacitor, and secondary side series tuning inductor, respectively. In Fig. 8, voltage stress indicates the voltage across these components, and current stress indicates the current through these components.

As seen from the tradeoff analysis in Figs. 8–10, the delta-delta connection of the three-phase WPT system results in the best (least) current stresses in the passive components for series-series, *LCC-LCC*, and *LCC-series* resonant tuning circuits; however, the total current stresses in *LCC-LCC* tuning parameters are higher than stresses in the series-series and *LCC-series* tuning circuits. The series-series tuning configuration has relatively lower voltage stresses considering all of the

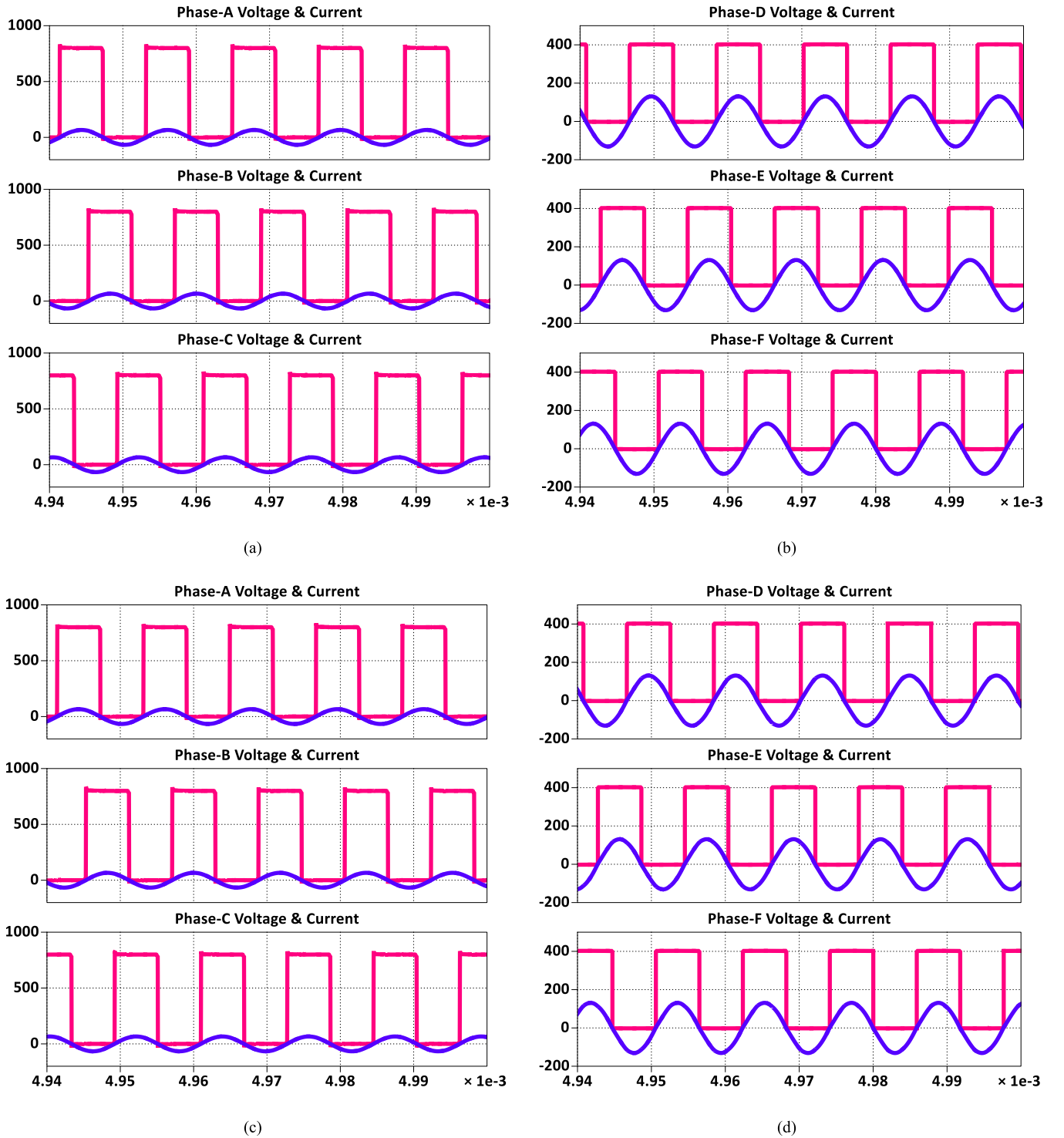


Fig. 11. Voltage/current stresses of a three-phase converter with series-series tuning: (a) resonant tank voltage and current in the primary side for three-phase coupler star/star ($Y-Y$) connection, (b) resonant tank voltage and current in the secondary side for three-phase coupler star-star ($Y-Y$) connection, (c) resonant tank voltage and current in the primary side for three-phase coupler delta/star (Δ/Y) connection, and (d) resonant tank voltage and current in the secondary side for three-phase coupler delta/star ($\Delta-Y$) connection.

components; however, when the three-phase coupler is configured in a delta-delta connection, each component's voltage stress is higher than those of the $LCC-LCC$ and LCC -series (primary) tuning compensations. The voltage stresses across the passive components are, however, in relatively acceptable voltage ranges in series-series compensation. As seen

from the results, the series-series tuning circuit shows better performance considering voltage and current stresses in each component.

LCC tuning network can, however, be designed to adjust to achieve a desired voltage and current gain in the system and provides additional design flexibility compared to the

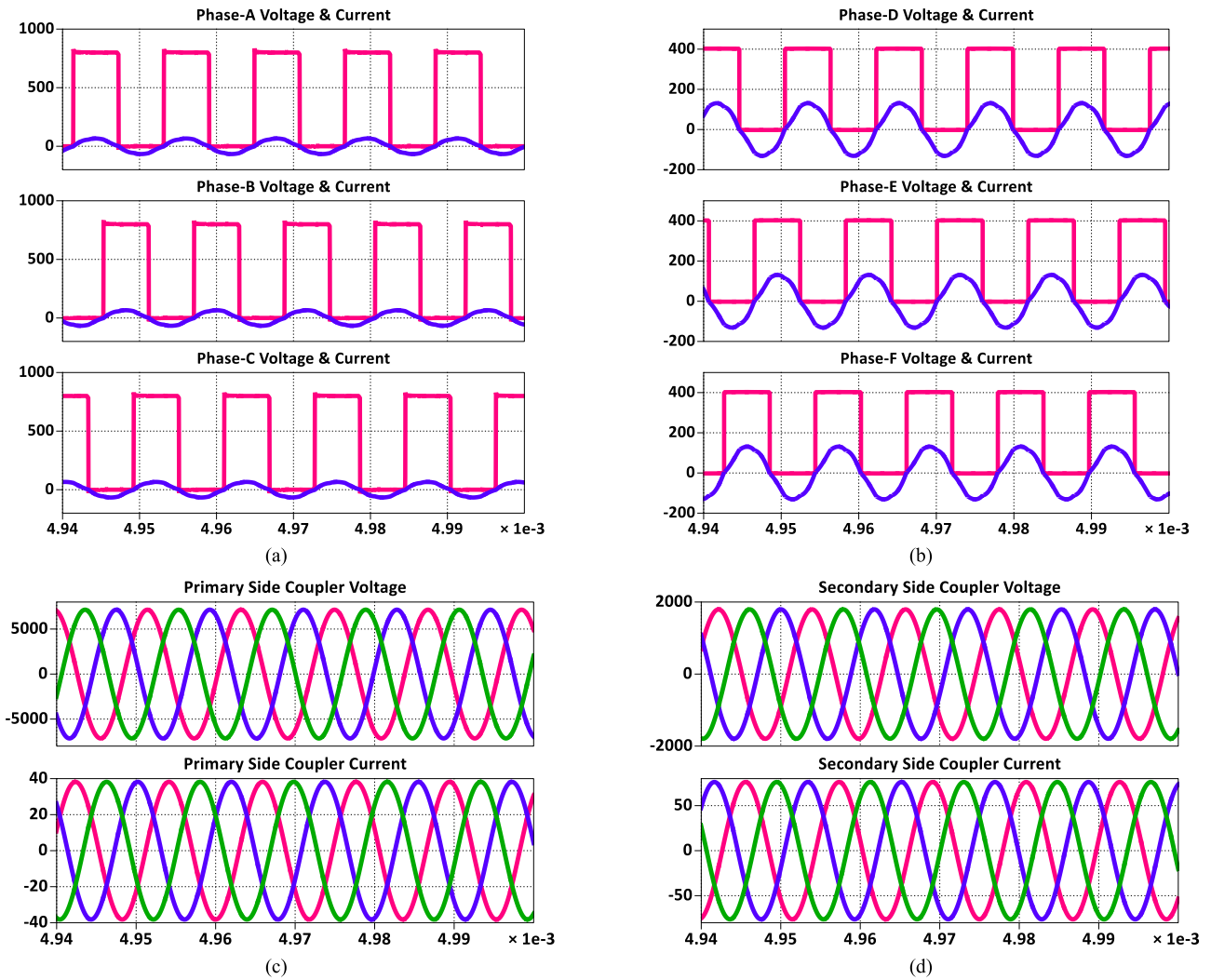


Fig. 12. Voltage/current stresses of a three-phase converter with *LCC-LCC* tuning for three-phase coupler delta-delta (Δ/Δ) connection: (a) resonant tank voltage and current in the primary side, (b) resonant tank voltage and current in the secondary side, (c) three-phase coupler voltage and current stresses in primary side, and (d) three-phase coupler voltage and current stresses in secondary side.

series-series tuning. Although *LCC-LCC* tuning provides the same current stress as series tuning in the three-phase series tuning capacitor, the series inductor connected to the switch node and parallel capacitor in the primary and secondary side terminals has relatively higher current stresses. The voltage stresses are, however, relatively lower compared with other passive components. Considering these features in *LCC-LCC* compensation, *LCC-series* tuning can be another option providing less current stress in the secondary side, similar to series-series tuning. The compensation circuit, however, still suffers from high current stresses in the primary side components. The advantage of *LCC-series* tuning is the current stress in the secondary side is relatively lower compared with *LCC-LCC*. According to these results, in terms of component stress levels, the series-series tuned delta-delta connection configuration performs better with lower total voltage and current stresses. On the other hand, it should be noted that primary-side *LCC* tuning provides additional benefits. With *LCC* tuning, primary-phase windings can act as load and coupling factor independent constant current sources, which

simplify the control and communications requirements. The *LCC-series* tuning circuit, moreover, has advantages that make it a suitable option for EV charger systems.

The component voltage and current stress comparisons are wye and delta connected three-phase coupler combinations are presented in Figs. 9 and 10.

These comparisons consider the voltage and current stresses across the passive components in the compensation network and the coils. The cost, size, weight, volume, temperature, and aging effects are the other parameters that can be considered in future comparisons. The simulation results for the parallel combination of 300 kW WPT EV charger are provided considering 50 kW each parallel converter with different coupler connection combinations, as shown in Figs. 11 and 12. The simulated system voltage and current waveforms are given for 800 V input and 400 V output at 50 kW for each parallel connection of the three-phase converter. Fig. 11(a) and (b) demonstrate the primary and secondary side resonant tank voltage and current results for the star-star ($Y-Y$) connection of the coupler system, respectively.

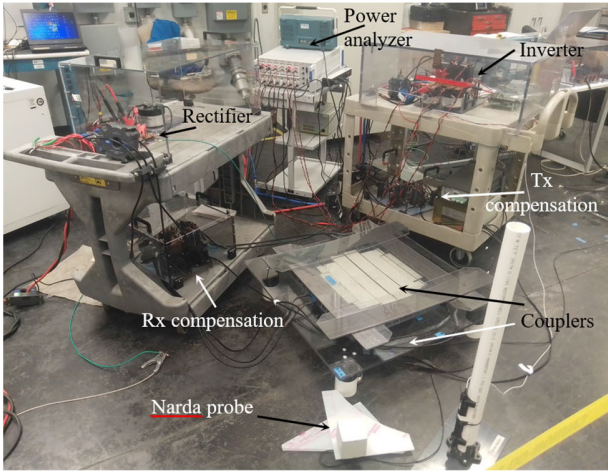


Fig. 13. Experimental test setup of the 50-kW polyphase WPT system, Tx and Rx couplers, compensation networks, inverter, and rectifier.

In addition to these figures, Fig. 11(c) and (d) show the resonant tank voltage and current waveforms for the delta-star ($\Delta - Y$) connection of the coupler system on the primary and secondary sides, respectively. As seen in Fig. 11, the resonant current is purely sinusoidal and close to the resonant point; each phase current is around 47 A in the primary side resonant tank and 93 A in the secondary side resonant tank. The delta configuration of the coupler, however, results in less current, around 27 A in the primary side coupler phases and around 54 A in the secondary side coupler phases. System simulation results with *LCC-LCC* resonant tuning network configuration are illustrated in Fig. 12. The system resonant tank voltage and current waveforms are depicted from the primary and secondary side resonant tank in Fig. 12(a) and (b), respectively. Fig. 12(c) and (d) show the primary and secondary side coil phase currents, respectively. As seen from Fig. 12, the resonant tank current values are similar to the series-series tuned coupler system with 47 A on the primary side and 93 A on the secondary side; however, the coil currents delta connected the system to around 27 A per phase in the primary side and 54 A per phase in the secondary side coupler windings.

B. Experimental Setup

The overall experimental test setup is shown in Fig. 13 along with the subsystems, components, and instrumentation devices. For the experimental validations, a three-phase inverter and rectifier and two-layer three-phase coupling coils are prototyped at the authors' laboratory. Coils use two layers with spatially shifted 180° and three-phase windings that are spatially shifted 120° from each other. The detailed view of the coupling coils is given in Fig. 14, where (a) shows the finite element analysis (FEA) model, (b) shows the hardware prototype, and (c) shows the receiver (Rx) and Tx coils on top of each other for the laboratory tests.

The coupler shown in Fig. 14 has bounding box dimensions of 47.1×54.4 cm, and the design is rated for 55 kW peak power rating primarily limited by the current rating of the 6 AWG Litz wire used in the prototype. The total mass of

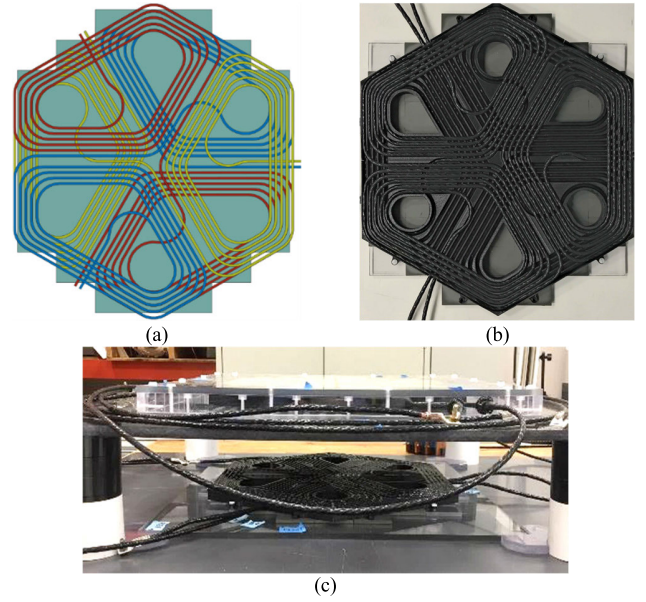


Fig. 14. Two-layer polyphase coupler: (a) FEA model of the coupler, (b) hardware prototype of the coupler, and (c) identical Tx and Rx placed on top of each other.

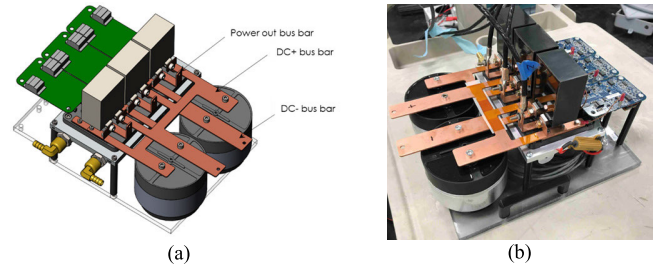


Fig. 15. (a) Design model of the three-phase inverter and (b) prototype hardware development.

one of the couplers is 13.7 kg, including ferrite, Litz wire, and 3-D printed wire guides. For the experiments, the ground clearance of the secondary-side coupler (distance from the surface of the Rx pad to the ground) is 180 mm, whereas the magnetic airgap is 170 mm. The wire is a six AWG Type-2 Litz wire with 38 AWG wire strands manufactured by New England Wire. The backplate ferrite was made with PLT64/50/5 ferrite tiles with 5 mm thickness and 3C94 material manufactured by Ferroxcube. The inductance matrix of the polyphase WPT system is provided in the following equation:

$$L = \begin{bmatrix} 31.6 & -9.8 & -9.8 & 5.3 & -1.2 & -3.3 \\ -9.8 & 32.2 & -10.2 & -3.7 & 5.2 & -1.3 \\ -9.8 & -10.2 & 32.3 & -1.5 & -3.5 & 5.1 \\ 5.3 & -3.7 & -1.5 & 31.1 & -10.0 & -9.8 \\ -1.2 & 5.2 & -3.5 & -10.0 & 31.3 & -9.9 \\ 3.3 & 1.3 & -5.1 & -9.8 & -9.9 & 31.6 \end{bmatrix} \mu\text{H}. \quad (45)$$

The three-phase inverter design in this study uses three 1200 V/325 A SiC MOSFET phase-leg modules (CAS325M12HM2) and gate drivers (CGD15HB62LP) manufactured by Wolfspeed/CREE. The heatsink is a liquid-cooled cold plate (CP3009) from Microcool. The inverter assembly uses ORNL custom designed dc bus bars with two 600 μF

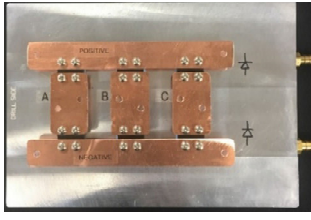
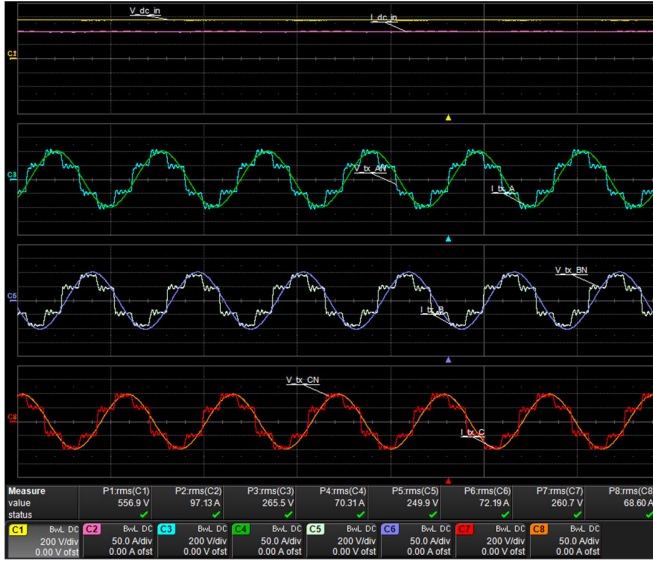
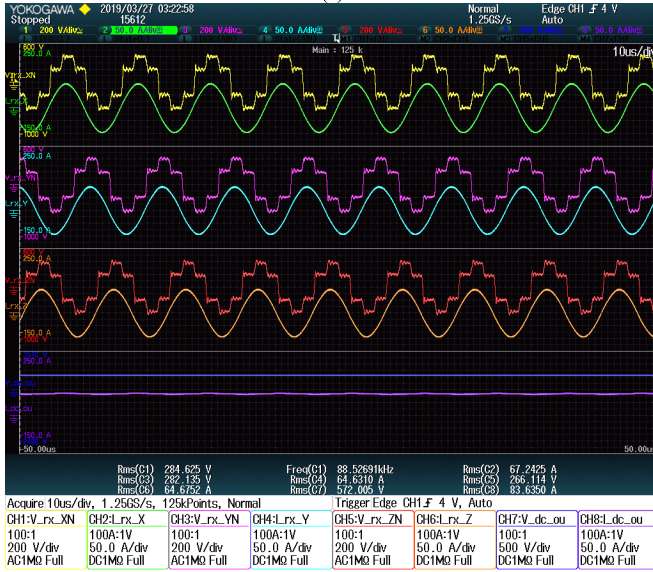


Fig. 16. Hardware prototype of the three-phase rectifier with GeneSiC diodes.



(a)



(b)

Fig. 17. Series-series compensation $Y-Y/Y-Y$: (a) dc-input and inverter's output voltages and currents and (b) rectifier's input and dc output voltages and currents.

dc-link capacitors (947D601K901DCRSN) and a pair of $3\ \mu\text{F}$ decoupling capacitors (SCD305K122C3Z25-F), one on each module. The inverter gate driver signals are produced from a TMS320F28335 digital signal processor (DSP), and the deadtime on the phase-leg module switches is set to 600 ns. The operating frequency of the inverter is selected to achieve zero-voltage switching (ZVS) around 85 kHz at full load for

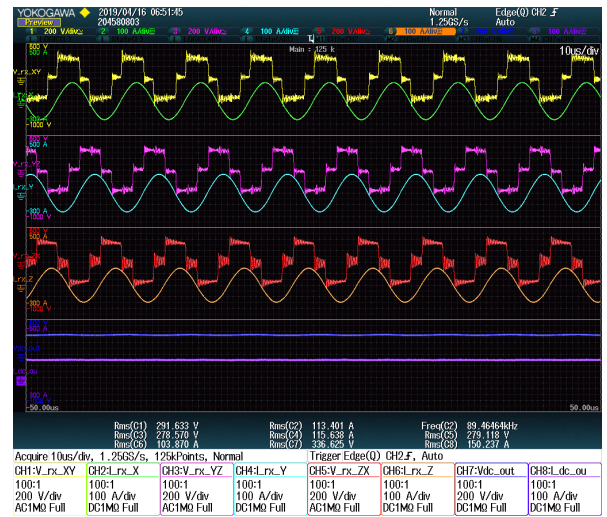
TABLE VII

RESONANT TUNING NETWORK PARAMETERS FOR DIFFERENT TUNING AND CONNECTION CONFIGURATIONS FOR EXPERIMENTAL VALIDATIONS

Configuration	Component	Parameter value	V_{rms}/I_{rms} rating	Assembly /Structure
Series-series tuning, $Y-Y/Y-Y$	C' [nF]	90.7	1490 V_{rms}	$(0.25 \parallel 0.25) - (0.25 \parallel 0.33)$
Series-series tuning, $\Delta-Y/\Delta-Y$	C' [nF]	267.2	2206 V_{rms}	0.25-0.25-0.33
LCC-LCC tuning, $\Delta-\Delta/\Delta-\Delta$	$L_{f\alpha}, L_{f\beta}, L_{f\gamma}$ [μH]	5.2, 4.5, 4.2	80 A_{rms}	Coil: 5 turns with bifilar AWG10 Litz wire Core: EE 80/38/20
	C_f [μF]	0.255	900 V_{rms}	CELEM CSP 120/200 0.17 $\parallel$ 0.085
	C_s [μF]	0.34	900 V_{rms}	CELEM CSP 120/200 0.17 $\parallel$ 0.17



(a)



(b)

Fig. 18. Series-series compensation $\Delta-Y/\Delta-Y$: (a) inverter output voltages and currents and (b) rectifier input voltages and currents.

the aligned coupler condition. The 3-D rendered image of the inverter and the hardware prototype developed at the authors' laboratory are given in Fig. 15.

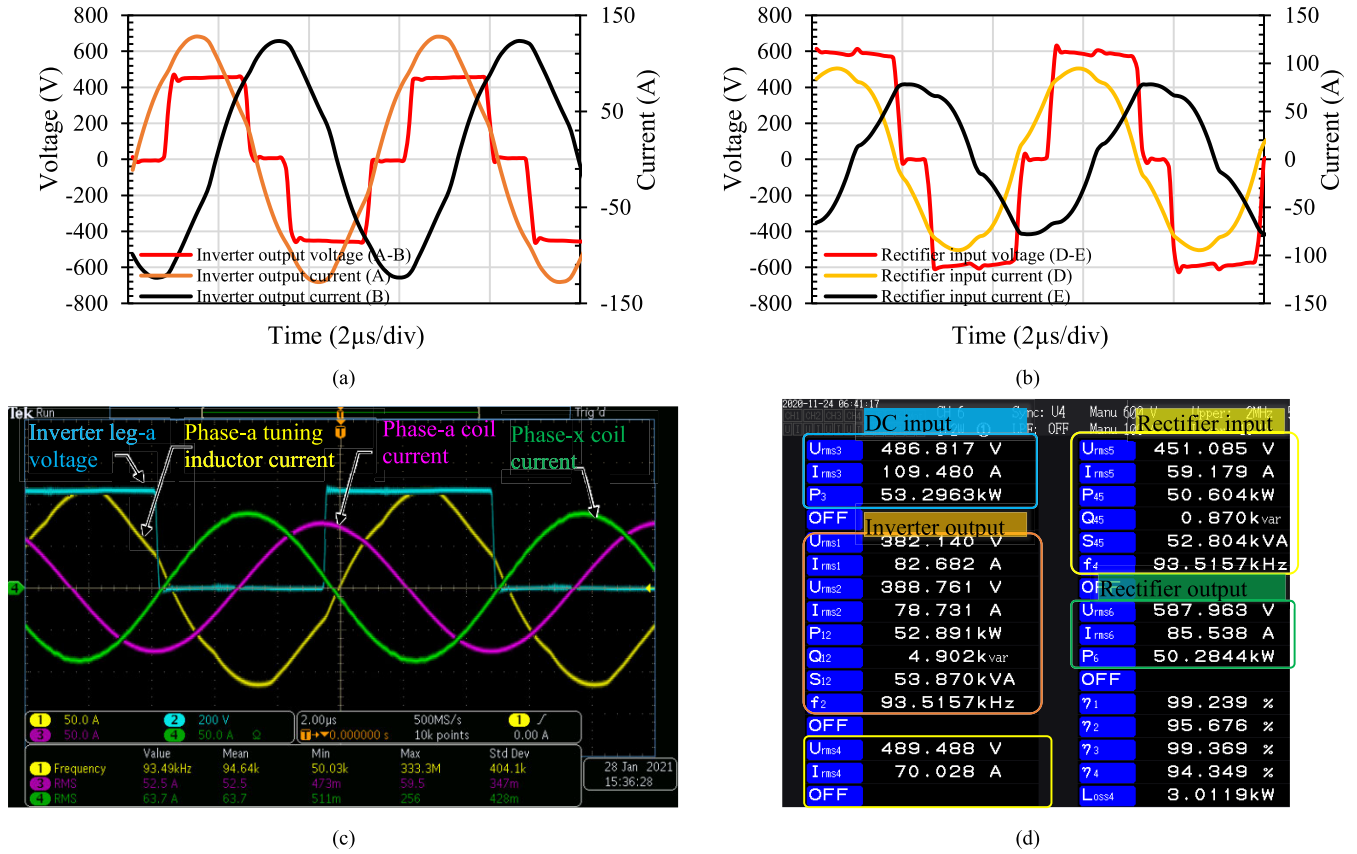


Fig. 19. *LCC-LCC* compensation test results: (a) inverter output voltage and currents, (b) rectifier input voltage and currents, (c) primary and secondary coil currents, and (d) power analyzer results for input and output voltages, currents, powers, and the efficiency.

For the rectifier for series-series tuned configuration testing, a separate identical inverter unit is used as the rectifier by keeping the switch gate signals in off mode. For the *LCC-LCC* tests, the rectifier is replaced with a three-phase rectifier using the 1200 V/200 A rated GB2X100MPS12-227 SiC junction barrier Schottky diodes manufactured by GeneSiC in SOT-227 packages with custom fabricated phase input and dc output bus bars as shown in Fig. 16. Each package has two parallel diodes for current sharing that reduces losses. This simpler diode-based rectifier's operation was found to be very similar to the SiC MOSFET-based rectifier that uses the internal diodes of the MOSFETs, but the diode bridge rectifier certainly reduces the cost, weight, and volume on the secondary side. The resonant tuning parameters, depending on the resonant tuning and connection configurations of the components and the coil windings, are presented in Table VII, which also presents assembly and construction details and component stresses in terms of voltage and current rms values.

C. Experimental Results

This section presents the experimental results of the studied 50-kW polyphase wireless charging system under aligned conditions for three different compensation topologies: $Y-Y/Y-Y$ and $\Delta-Y/\Delta-Y$ configured series-series compensation and $\Delta-\Delta/\Delta-\Delta$ configured *LCC-LCC* compensation. While the system was designed for nominal 85 kHz

resonance frequency, the operating frequency of the three compensation topologies varied between 88.5 and 93.5 kHz to achieve the ZVS operation of the inverter and maximum dc-to-dc efficiency.

The experimental results of the $Y-Y/Y-Y$ configured series-series connected system with aligned couplers are shown in Fig. 17. Output voltages and currents of the inverter are shown in Fig. 17(a), and input voltages and currents of the rectifier are shown in Fig. 17(b). Two oscilloscopes were used to monitor the large number of currents and voltages from the Tx and Rx sides. The inverter was operated at 88.5 kHz to achieve ZVS and maximum efficiency. As seen in Fig. 17, the dc-to-dc gain was near unity with 557 V input and 572 V output dc voltage.

The test results of the $\Delta-Y/\Delta-Y$ configured series-series connected systems are shown in Fig. 18. Output voltages and currents of the inverter are shown in Fig. 18(a), and input voltages and currents of the rectifier are shown in Fig. 18(b). The inverter was operated at 89.46 kHz, compared with the 88.5 kHz operating frequency of the $Y-Y/Y-Y$ system. The experimental voltage and current waveforms and efficiencies of the $\Delta-\Delta/\Delta-\Delta$ connected *LCC-LCC* compensation are shown in Fig. 19. The input and output voltages of the inverter and rectifier were observed through a high-frequency HIOKI PW6001 power analyzer, and the primary and secondary coil currents were observed using a Tektronix MDO3000 oscilloscope.

The output voltage and currents of the inverter are shown in Fig. 19(a), and the input currents and voltage of the rectifier are shown in Fig. 19(b). Fig. 19(c) shows the primary and secondary coil currents, both of which are highly sinusoidal; the phase-A coil current is 90° phase apart from the phase-A inverter voltage. The inverter was operated at 93.5 kHz to achieve the ZVS and maximum efficiency. The efficiency of the inverter (η_1), resonant tank (η_2), rectifier (η_3), and overall dc-to-dc system (η_4) are shown in Fig. 19(d). The results indicate that while the inverter and rectifier are highly efficient (>99%), the resonant tank can be further optimized to enhance the overall efficiency.

VI. CONCLUSION

This study focuses on wireless fast charging systems that are accomplished through polyphase wireless coupling coils. After the review of grid connected PFC rectifier topologies, series and parallel connected high-power charging systems that are needed to achieve high power levels greater than 300 kW are discussed. Three-phase electromagnetic coupler connection configurations are discussed next, along with the resonant tuning component connection configurations and their sizing and component value equations derived for each connection configuration. These star and delta connection configuration options for the coupler windings and tuning networks are compared in a tradeoff analysis with their impact on component voltage and current stresses. Simulation results are presented on most practical configurations for series-series and *LCC-LCC* tuned systems. Experimental results were presented for $Y - Y/Y - Y$ connected and $\Delta - Y/\Delta - Y$ connected series-series tuned polyphase WPT systems as well as $\Delta - \Delta/\Delta - \Delta$ connected *LCC-LCC* tuned polyphase systems to validate system operation with tuning component sizing dependent on the connection configuration. **Experimental results show about 94.35% dc-to-dc efficiency at 50 kW output power transfer.** This study can serve as a design guideline for all the possible polyphase resonant tuning network and phase winding connection configurations for *LCC-LCC* and series-series tuned polyphase WPT systems.

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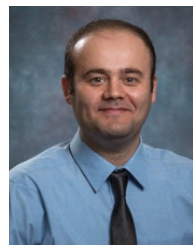
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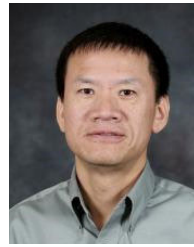
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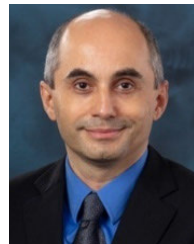
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